

TRILINEAR KLOOSTERMAN FRACTIONS I: PARTIALLY FIXED MODULI AND UNBALANCED CONVOLUTIONS

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ABSTRACT. In this paper, we improve on Fouvry and Radziwiłł's results on unbalanced convolutions. In particular, we find that if (α_m) and (β_n) are sequences supported on $m \sim M$ and $n \sim M$ where β_n is equidistributed for small moduli, then

$$\sum_{q \sim Q} \left| \sum_{\substack{n \sim N, m \sim M \\ mn \equiv a \pmod{q}}} \alpha_m \beta_n - \frac{1}{\phi(q)} \sum_{\substack{n \sim N, m \sim M \\ (mn, q) = 1}} \alpha_m \beta_n \right| \ll \frac{X}{\log^A X},$$

as long as $\exp((\log x)^\varepsilon) \leq N \leq Q^{-11/12} X^{17/36 - \varepsilon}$ with $Q \leq X^{1/2 + 1/66 - \delta}$, along with wider bounds for N if $Q \leq X^{\frac{45}{89} - \varepsilon}$. The former improves the allowable range of N , while the latter improves the range of Q . To prove these new bounds, we improve Bettin and Chandee's famous result on trilinear forms with Kloosterman fractions in the case where the denominator has a fixed factor.

1. INTRODUCTION

A common question in number theory is this: given an arithmetic function f and an integer $a \neq 0$, what is the largest $\delta > 0$ for which $Q \leq X^{\frac{1}{2} + \delta}$ implies that

$$\sum_{q \sim Q} \left| \sum_{\substack{n \sim X \\ n \equiv a \pmod{q}}} f(n) - \frac{1}{\phi(q)} \sum_{\substack{n \sim X \\ (n, q) = 1}} f(n) \right| \ll \frac{X}{\log^A X}$$

for some (or possibly for every) $A > 0$?

Often - such as in the case of the primes - such questions can be expressed in terms of convolutions. For $m \sim M$ and $n \sim N$ with $MN = X$, define α_m and β_n to be sequences of complex numbers. We then wish to find a δ such that for $Q \leq X^{1/2 + \delta}$,

$$(1) \quad \sum_{\substack{q \sim Q \\ (a, q) = 1}} \left| \sum_{\substack{n \sim N, m \sim M \\ mn \equiv a \pmod{q}}} \alpha_m \beta_n - \frac{1}{\phi(q)} \sum_{\substack{n \sim N, m \sim M \\ (mn, q) = 1}} \alpha_m \beta_n \right| \ll \frac{X}{\log^A X}.$$

The cases where $\delta < 0$ follow from the generalized version of the Bombieri-Vinogradov Theorem. Cases where $\delta > 0$, however, are often much more difficult. For instance, in the case of the primes, this is the Elliot-Halberstam conjecture, and thus at present there is no fixed $\delta > 0$ for which (1) can be shown to hold if $\alpha_m = \mu(m)$ and $\beta_n = \log n$.

In order for it to be possible for (1) to hold, it is generally required that at least one of the two functions is equidistributed for small moduli. We call such equidistribution the *Siegel-Walfisz condition*, defined as follows:

Definition 1. An arithmetic function β_n is defined to be Siegel-Walfisz if there exists an integer $k > 0$ such that for any fixed $A > 0$, uniformly in $x \geq 2$, $q > |a| \geq 1$, $r \geq 1$, and $(a, q) = 1$, we have

$$\sum_{\substack{x \leq n \leq 2x \\ n \equiv a \pmod{q} \\ (n, r) = 1}} \beta_n = \frac{1}{\phi(q)} \sum_{\substack{x \leq n \leq 2x \\ (n, qr) = 1}} \beta_n + O(x(\log x)^{-A} \tau_k(r)).$$

Here, τ_k denotes the k -fold divisor function.

In 2018, Fouvry and Radziwiłł showed that (1) holds for small positive values of $\delta > 0$ as long as N is fairly small relative to M . In particular, Corollary 1.1 of [FR] was the following:

Corollary 1.1. Let $k > 0$ and $\varepsilon > 0$ be given. Let α_m and β_n be two sequences of complex numbers such that

$$|\alpha_m| \leq \tau_k(m) \quad \text{and} \quad |\beta_n| \leq \tau_k(n)$$

for every $m, n > 1$. Suppose that β_n satisfies the Siegel-Walfisz condition. Then for every $A > 0$, uniformly for $M, N \geq 2$ with

$$\frac{MN}{2} \leq X \leq 4MN,$$

we have

$$\sum_{\substack{Q \leq q \leq 2Q \\ (q, a) = 1}} \left| \sum_{\substack{X < mn \leq 2X \\ mn \equiv a \pmod{q}}} \alpha_m \beta_n - \frac{1}{\phi(q)} \sum_{\substack{X < mn \leq 2X \\ (mn, q) = 1}} \alpha_m \beta_n \right| \ll X(\log X)^{-A},$$

provided that one of the following three conditions holds:

- (i) $\exp((\log X)^\varepsilon) \leq N \leq Q^{-11/12} X^{17/36-\varepsilon}$ and $1 \leq |a| \leq X/12$;
- (ii) $\exp((\log X)^\varepsilon) \leq N \leq X^{7/90-\varepsilon}$, $Q \leq X^{53/105-\varepsilon}$, and $1 \leq |a| \leq X/12$;
- (iii) $\exp((\log X)^\varepsilon) \leq N \leq X^{101/630-\varepsilon}$, $Q \leq X^{53/105-\varepsilon}$, and $1 \leq |a| \leq (X/4)^{\varepsilon/1000}$.

A key step in this result involves a bound for trilinear forms with Kloosterman fractions discovered by Bettin and Chandee [BC]. Define

$$\mathcal{B}(M, N, A) = \sum_{a \sim A} \sum_{\substack{m \sim M \\ (m, n) = 1}} \sum_{n \sim N} \alpha_m \beta_n \nu_a e\left(\vartheta \frac{a \overline{m}}{n}\right),$$

where ϑ is a nonzero integer.

In [BC], the authors then prove the following:

Theorem 1.2.

$$\begin{aligned} \mathcal{B}(M, N, A) &\ll \|\alpha\| \|\beta\| \|\nu\| \left(1 + \frac{|\vartheta| A}{MN}\right)^{\frac{1}{2}} \\ &\quad \times \left((AMN)^{\frac{7}{20}+\varepsilon} (M+N)^{\frac{1}{4}} + (AMN)^{\frac{3}{8}+\varepsilon} (AN+AM)^{\frac{1}{8}}\right). \end{aligned}$$

2. NEW IDEAS

In this paper, we alter the Bettin-Chandee result to make it slightly more helpful for the Fouvry-Radziwiłł result.

We recall that the [FR] work considers functions $g(m)$ and $h(n)$ with $m \sim M$ and $n \sim N$ where MN is roughly of size X but $\exp((\log X)^\theta) \leq N \leq X^{1/72-\epsilon}$ (i.e. N is much smaller than M). When the authors of [FR] then apply the result of [BC], they have a sum of the form

$$(2) \quad \sum_{a \sim A} \sum_{\substack{b \sim Q, q \sim Q \\ (b, Rq)=1}} f_1(b) f_2(q) f_3(a) e\left(\vartheta \frac{a\bar{b}}{Rq}\right)$$

for a parameter $R \leq 2N$. Translating their work into the language of Bettin and Chandee, they then define their function α_n to be such that n lies on intervals of the form $[RQ, 2RQ]$ with the caveat that $\alpha_n = 0$ if $R \nmid n$. This allows them to apply Theorem 1.2, but at the slight cost of having to replace Q with NQ in the bound.

However, the original proof of Bettin and Chandee (and the preceding work of Duke, Friedlander, and Iwaniec) already has a useful mechanism that one could exploit to find a better bound for something like (2). In [BC], the authors define

$$(3) \quad \mathcal{C}_1(M, N, A) = \sum_{m \sim M} \left| \sum_{\substack{a \sim A, n \sim N \\ (m, n)=1}} \beta_n \nu_a e\left(\vartheta \frac{a\bar{m}}{n}\right) \right|^2$$

and then introduce the so-called "complementary divisor" b , considering

$$\mathcal{C}_b(M, N, A; \beta, \nu) = \sum_{\substack{m \in \mathcal{M} \\ (m, b)=1}} \left| \sum_{\substack{a \in \mathcal{A}, n \in \mathcal{N} \\ (m, n)=1}} \beta_n \nu_a e\left(\theta \frac{a\bar{m}}{nb}\right) \right|^2.$$

The authors then relate $\mathcal{C}_1$ and $\mathcal{C}_b$ back to $\mathcal{B}$, bounding the latter via estimates on $\mathcal{C}_1$ and $\mathcal{C}_b$ as well as Cauchy-Schwarz.

If one were to consider instead

$$\sum_{m \sim M} \left| \sum_{\substack{a \sim A, n \sim N \\ (m, n)=1}} \beta_n \nu_a e\left(\vartheta \frac{a\bar{m}}{Rn}\right) \right|^2$$

and

$$\sum_{m \sim M} \left| \sum_{\substack{a \sim A, n \sim N \\ (m, n)=1}} \beta_n \nu_a e\left(\vartheta \frac{a\bar{m}}{bRn}\right) \right|^2,$$

the sum now already has a built-in complementary divisor in R , and hence one could use the same process the authors used to bound $\mathcal{C}_b$ to find bounds for these

expressions as well. Applying this process and Cauchy-Schwarz, we can then give a bound for

$$\mathcal{B}(M, N, A; R) := \sum_{a \in \mathcal{A}} \sum_{\substack{m \in \mathcal{M}, n \in \mathcal{N} \\ (m, nR)=1}} \alpha_m \beta_n \nu_a e \left(\theta \frac{a\overline{m}}{nR} \right)$$

in the same way as before. This is the mechanism in the Bettin-Chandee proof that we use to find improvements on the Fouvry-Radziwill bound.

In particular, we will prove the following version of Bettin and Chandee's result:

Theorem 2.1. *Let $M \ll N^2$ and $R \ll M^A$ for some large A . Then*

$$\begin{aligned} \mathcal{B}(M, N, A; R) &\ll M^\epsilon \|\alpha\| \|\nu\| \|\beta\| (AMN)^{\frac{1}{2}} R^{\frac{1}{4}} \left(1 + \frac{|\vartheta|A}{MN} \right)^{\frac{1}{4}} \\ &\times \left(\frac{1}{N^{\frac{1}{8}}} + \frac{R^{\frac{1}{8}} N^{\frac{1}{8}}}{M^{\frac{1}{4}}} + \frac{M^{\frac{1}{10}}}{R^{\frac{3}{20}} A^{\frac{1}{20}} N^{\frac{3}{20}}} + \frac{N^{\frac{3}{20}}}{A^{\frac{3}{20}} M^{\frac{1}{5}}} + \frac{N^{\frac{3}{8}}}{M^{\frac{1}{2}}} \right) \end{aligned}$$

When $R = 1$, this is (7.2) of [BC]. However, it is clear that when R is much larger than 1, this theorem will yield a smaller bound than would simply substituting NR for N in Theorem 1.2.

Armed with this theorem, we record the following improvement of Corollary 1.1:

Corollary 2.2. *Let $k > 0$ and $\varepsilon > 0$ be given. Let α_m and β_n be k -fold divisor-bounded sequences of complex numbers as in Corollary 1.1 with β_n satisfying the Siegel-Walfisz condition. Then for every $A > 0$, uniformly for $M, N \geq 2$ with*

$$\frac{MN}{2} \leq X \leq 4MN,$$

we have

$$(4) \quad \sum_{\substack{Q \leq q \leq 2Q \\ (q, a)=1}} \left| \sum_{\substack{X < mn \leq 2X \\ mn \equiv a \pmod{q}}} \alpha_m \beta_n - \frac{1}{\varphi(q)} \sum_{\substack{X < mn \leq 2X \\ (mn, q)=1}} \alpha_m \beta_n \right| \ll_A x (\log X)^{-A},$$

provided that one of the following three conditions holds:

- (i) $\exp((\log X)^\varepsilon) \leq N \leq Q^{-33/28} X^{17/28-\varepsilon}$ and $1 \leq |a| \leq X/12$;
- (ii) $\exp((\log X)^\varepsilon) \leq N \leq X^{7/90-\varepsilon}$, $Q \leq X^{45/89-\varepsilon}$, and $1 \leq |a| \leq X/12$;
- (iii) $\exp((\log X)^\varepsilon) \leq N \leq X^{101/630-\varepsilon}$, $Q \leq X^{45/89-\varepsilon}$, and $1 \leq |a| \leq (X/4)^{\varepsilon/1000}$.

Note that in the extremal case where N is very small, the bound in (i) gives $Q \leq X^{1/2+1/66-\varepsilon}$, which is the same as in [FR]. This is of course because the savings are entirely dependent on the size of N , and thus if N is very small we have no appreciable savings. On the other hand, if $Q = X^{\frac{1}{2}+\varepsilon}$ then (i) of our corollary gives $N \leq X^{1/56-\varepsilon'}$ for some ε' depending on ε , which is an improvement on the $N \leq X^{1/72-\varepsilon'}$ from Corollary 1.1.

In their paper, Fouvry and Radziwill state Corollary 1.1 as the consequence of an estimate involving Linnik's dispersion. Define

$$E(\alpha, \beta, M, N, q, a) := \sum_{\substack{m \sim M \\ n \sim N \\ mn \equiv a \pmod{q}}} \alpha_m \beta_n - \frac{1}{\varphi(q)} \sum_{\substack{m \sim M \\ n \sim N \\ (mn, q)=1}} \alpha_m \beta_n,$$

and define

$$\Delta(\alpha, \beta, M, N, Q, a) := \sum_{\substack{q \sim Q \\ (q, a)=1}} |E(\alpha, \beta, M, N, q, a)|.$$

For the contribution of the small moduli to $\Delta(\alpha, \beta, M, N, Q, a)$, we have

$$\mathcal{E}^*(\beta, N, Q) := \sum_{\delta} \sum_{v \sim Q/\delta} \sum_{\substack{\delta' \pmod{\delta} \\ (\delta', \delta)=1}} |E(\beta, N, \delta, \delta'; v)|^2.$$

Note that if β is Siegel-Walfisz and satisfies $|\beta_n| \leq \tau_k(n)$ for some $k > 0$, and if $N > Q^\varepsilon$, then

$$\mathcal{E}^*(\beta, N, Q) \ll_A N^2 Q (\log N)^{-A}.$$

In this paper, we will prove the following:

Theorem 2.3. *Let k, ε, α_m , and β_n be as in the previous corollary. Let $k \geq 1$ be an integer and let $\varepsilon > 0$ be given.*

$$M > Q(MN)^\varepsilon, \quad M > N > D^{10}.$$

Then, for all integers $1 \leq |a| \leq X/3$, we have

$$|\Delta(\alpha, \beta, M, N, Q, a)| \ll \|\alpha\| \left(MQ^{-1} \mathcal{E}^*(\beta, N, Q) + (\log X)^\kappa N^2 Q + (\log X)^\kappa N^2 D^{-\frac{1}{2}} M + D^C X^\epsilon \left(Q^{\frac{15}{8}} N^{\frac{11}{4}} + M^{\frac{3}{20}} Q^{\frac{33}{20}} N^{\frac{51}{20}} \right) \right)^{\frac{1}{2}}.$$

In [FR], the last term is

$$D^C X^\epsilon \left(Q^{\frac{15}{8}} N^{\frac{23}{8}} + M^{\frac{3}{20}} Q^{\frac{33}{20}} N^{\frac{59}{20}} \right).$$

Hence we have a savings of $N^{\frac{1}{8}}$ in the first term and $N^{\frac{2}{5}}$ in the second.

We note that Theorem 2.3 also gives an improvement of [FR, Corollary 1.5].

Corollary 2.4. *Let $\varepsilon > 0$ and $k > 0$ be given. Let λ_d be a sequence of complex numbers with $|\lambda_d| \leq \tau_k(d)$. Then*

$$\sum_{\substack{q \sim Q \\ (q, a)=1}} \left| \sum_{\substack{X < n \leq 2X \\ n \equiv a \pmod{q}}} \sum_{\substack{d|n \\ d \leq z}} \lambda_d - \frac{1}{\varphi(q)} \sum_{\substack{X < n \leq 2X \\ (n, q)=1}} \sum_{\substack{d|n \\ d \leq z}} \lambda_d \right| \ll_A \frac{X}{(\log X)^A},$$

provided that one of the following three conditions holds:

- (i) $X \geq 12$, $z \leq X^{45/89-\varepsilon}$, $X^{1-\varepsilon} > Q > X^{529/630+\varepsilon}$, and $1 \leq |a| \leq X^{\varepsilon/10000}$;
- (ii) $X \geq 12$, $z \leq X^{45/89-\varepsilon}$, $X^{1-\varepsilon} > Q > X^{83/90+\varepsilon}$, and $1 \leq |a| \leq X^{1-3\varepsilon}$;
- (iii) $X \geq 12$, $z \leq X^{1/2+\delta-\varepsilon}$, $X^{1-\varepsilon} > Q > X^{(55+66\delta)/56+\varepsilon}$, and $1 \leq |a| \leq X^{1-3\varepsilon}$ for any fixed $0 < \delta < \frac{1}{66}$. In this case the implicit constant in $\ll_A$ depends additionally on δ .

The proof of this corollary is essentially identical to that of Corollary 1.5 in [FR] with the changes made in our proof of Corollary 2.2 above. Hence, we omit this proof.

3. PROOF OF THEOREM 2.1: TRILINEAR FORMS WITH KLOOSTERMAN FRACTIONS AND A FIXED FACTOR IN THE DENOMINATOR

First, we give the proof for our improved version of Bettin and Chandee's result.

Let $\mathcal{N}_x$ denote the interval $[N/x, 2N/x]$. Recalling the definitions of $\mathcal{C}_1$ and C_b , we define $C_{b;R}$ as

$$C_{b;R}(M, N, A; \beta, \nu) = \sum_{\substack{m \in \mathcal{M} \\ (m, R)=1}} \left| \sum_{\substack{r|R \\ r \text{ square-free}}} \sum_{\substack{a \in \mathcal{A}, n \in \mathcal{N}_r \\ (mR, n)=1}} \beta_{nr} \nu_a e\left(\theta \frac{a\overline{m}}{nbrR}\right) \right|^2$$

Note that

$$(5) \quad \mathcal{B}(M, N, A; R) \ll \|\alpha\| \mathcal{C}_{1;R}(M, N, A; R; \beta, \nu)^{\frac{1}{2}}.$$

In (5.2), the authors of [BC] find that for any b ,

$$\begin{aligned} \mathcal{C}_b(M, N, A; \beta, \nu) &\ll \|\beta\|^2 \|\nu\|^2 M^\epsilon \left(1 + \frac{|\vartheta|A}{bMN}\right)^{\frac{1}{2}} \left(AM(bN)^{\frac{1}{2}} + b^{\frac{3}{4}} AM^{\frac{1}{2}} N^{\frac{5}{4}} + \frac{AM^{\frac{6}{5}} N^{\frac{1}{10}}}{b^{\frac{2}{5}}}\right. \\ &\quad \left.+ b^{\frac{1}{5}} A^{\frac{2}{5}} M^{\frac{6}{5}} N^{\frac{7}{10}} + b^{\frac{1}{2}} A^{\frac{7}{10}} M^{\frac{3}{5}} N^{\frac{13}{10}} + b^{\frac{1}{2}} AN^{\frac{7}{4}}\right). \end{aligned}$$

By our notation, we then have

$$\begin{aligned} \mathcal{C}_{1;R}(M, N, A; \beta, \nu) &= \sum_{\substack{m \in \mathcal{M} \\ (m, R)=1}} \left| \sum_{\substack{r|R \\ r \text{ square-free}}} \sum_{\substack{a \in \mathcal{A}, n \in \mathcal{N}_r \\ (mR, n)=1}} \beta_{nr} \nu_a e\left(\theta \frac{a\overline{m}}{nrR}\right) \right|^2 \\ &= \sum_{\substack{m \in \mathcal{M} \\ (m, R)=1}} \left| \sum_{\substack{b \text{ square-full} \\ (m, b)=1}} \sum_{\substack{r|R \\ r \text{ square-free}}} \sum_{\substack{a \in \mathcal{A}, n \in \mathcal{N}_{br} \\ (mbR, n)=1 \\ n \text{ square-free}}} \beta_{nbr} \nu_a e\left(\theta \frac{a\overline{m}}{nbrR}\right) \right|^2 \\ &\ll \tau(R) \sum_{\substack{b' \text{ square-full} \\ (m, b')=1}} (b')^{-\frac{1}{2}} \sum_{\substack{b \text{ square-full} \\ (m, b)=1}} b^{\frac{1}{2}} \sum_{\substack{r|R \\ r \text{ square-free}}} \sum_{\substack{m \in \mathcal{M} \\ (m, R)=1}} \left| \sum_{\substack{a \in \mathcal{A}, n \in \mathcal{N}_{br} \\ (mbR, n)=1 \\ n \text{ square-free}}} \beta_{nbr} \nu_a e\left(\theta \frac{a\overline{m}}{nbrR}\right) \right|^2 \\ &\ll \tau(R) \sum_{\substack{b \text{ square-full} \\ (m, b)=1}} b^{\frac{1}{2}} \sum_{\substack{r|R \\ r \text{ square-free}}} \mathcal{C}_{brR}(M, N/br, A; \beta_{br}, \nu), \end{aligned}$$

where $\beta_{br}(n) = \mu(n)^2 \beta_{brn}$. For the interval where $b > B$, we use the trivial bound that

$$\mathcal{C}_{brR}(M, N/br, A; \beta_{br}, \nu) \ll \|\nu\|^2 \frac{AMN}{br} \|\beta_{br}\|^2,$$

and hence these terms contribute

$$\ll \|\beta\|^2 \|\nu\|^2 \frac{ANM^{1+\epsilon}}{B^{\frac{1}{2}}}.$$

Taking $B = N^{\frac{1}{2}}$, we then have that the above is

$$\ll \|\beta\|^2 \|\nu\|^2 AN^{\frac{3}{4}} M^{1+\epsilon}.$$

For the terms with $b \leq B$, we apply our bound for $\mathcal{C}_b$ above, finding

$$\begin{aligned} \tau(R) & \sum_{\substack{b \text{ square-full} \\ (m,b)=1 \\ b \leq B}} b^{\frac{1}{2}} \sum_{\substack{r|R \\ r \text{ square-free}}} \mathcal{C}_{brR}(M, N/br, A; \beta_{br}, \nu) \\ & \ll M^\epsilon \|\nu\|^2 \sum_{\substack{b \text{ square-full} \\ (m,b)=1 \\ b \leq B}} \sum_{\substack{r|R \\ r \text{ square-free}}} \|\beta_{br}\|^2 \left(1 + \frac{|\vartheta|A}{MN}\right)^{\frac{1}{2}} \left(\frac{AMR^{\frac{1}{2}} N^{\frac{1}{2}}}{r^{\frac{1}{2}}} + \frac{R^{\frac{3}{4}} AM^{\frac{1}{2}} N^{\frac{5}{4}}}{b^{\frac{1}{2}} r^{\frac{1}{2}}} + \frac{AM^{\frac{6}{5}} N^{\frac{1}{10}}}{R^{\frac{2}{5}} b^{\frac{1}{2}} r^{\frac{1}{2}}} \right. \\ & \quad \left. + \frac{R^{\frac{1}{5}} A^{\frac{2}{5}} M^{\frac{6}{5}} N^{\frac{7}{10}}}{b^{\frac{1}{2}} r^{\frac{1}{2}}} + \frac{R^{\frac{1}{2}} A^{\frac{7}{10}} M^{\frac{3}{5}} N^{\frac{13}{10}}}{b^{\frac{4}{5}} r^{\frac{4}{5}}} + \frac{R^{\frac{1}{2}} AN^{\frac{7}{4}}}{b^{\frac{5}{4}} r^{\frac{5}{4}}} \right) \\ & \ll M^\epsilon \|\nu\|^2 \|\beta\|^2 \left(1 + \frac{|\vartheta|A}{MN}\right)^{\frac{1}{2}} \left(AMR^{\frac{1}{2}} N^{\frac{1}{2}} B^{\frac{1}{2}} + R^{\frac{3}{4}} AM^{\frac{1}{2}} N^{\frac{5}{4}} + \frac{AM^{\frac{6}{5}} N^{\frac{1}{10}}}{R^{\frac{2}{5}}} \right. \\ & \quad \left. + R^{\frac{1}{5}} A^{\frac{2}{5}} M^{\frac{6}{5}} N^{\frac{7}{10}} + R^{\frac{1}{2}} A^{\frac{7}{10}} M^{\frac{3}{5}} N^{\frac{13}{10}} + R^{\frac{1}{2}} AN^{\frac{7}{4}} \right). \end{aligned}$$

Letting $B = N^{\frac{1}{2}}$ as above, we use the fact that $M \ll N^2$, which allows us to write that $\frac{AM^{\frac{6}{5}} N^{\frac{1}{10}}}{R^{\frac{2}{5}}} \leq AMR^{\frac{1}{2}} N^{\frac{3}{4}}$. So

$$\begin{aligned} \mathcal{C}_{1;R}(M, N, A; \beta, \nu) & \ll M^\epsilon \|\nu\|^2 \|\beta\|^2 \left(1 + \frac{|\vartheta|A}{MN}\right)^{\frac{1}{2}} \left(AMR^{\frac{1}{2}} N^{\frac{3}{4}} + R^{\frac{3}{4}} AM^{\frac{1}{2}} N^{\frac{5}{4}} + R^{\frac{1}{5}} A^{\frac{2}{5}} M^{\frac{6}{5}} N^{\frac{7}{10}} + R^{\frac{1}{2}} A^{\frac{7}{10}} M^{\frac{3}{5}} N^{\frac{13}{10}} + R^{\frac{1}{2}} AN^{\frac{7}{4}} \right) \\ & = M^\epsilon \|\nu\|^2 \|\beta\|^2 AMN R^{\frac{1}{2}} \left(1 + \frac{|\vartheta|A}{MN}\right)^{\frac{1}{2}} \left(\frac{1}{N^{\frac{1}{4}}} + \frac{R^{\frac{1}{4}} N^{\frac{1}{4}}}{M^{\frac{1}{2}}} + \frac{M^{\frac{1}{5}}}{R^{\frac{3}{10}} A^{\frac{3}{5}} N^{\frac{3}{10}}} + \frac{N^{\frac{3}{10}}}{A^{\frac{3}{10}} M^{\frac{2}{5}}} + \frac{N^{\frac{3}{4}}}{M} \right). \end{aligned}$$

Thus by (5),

$\mathcal{B}(M, N, A; R)$

$$\begin{aligned} & \ll \|\alpha\| \|\nu\| \|\beta\| \left(1 + \frac{|\vartheta|A}{MN}\right)^{\frac{1}{4}} \\ & \quad \times \left(A^{\frac{1}{2}} M^{\frac{1}{2}} R^{\frac{1}{4}} N^{\frac{3}{8}} + R^{\frac{3}{8}} A^{\frac{1}{2}} M^{\frac{1}{4}} N^{\frac{5}{8}} + R^{\frac{1}{10}} A^{\frac{2}{10}} M^{\frac{3}{5}} N^{\frac{7}{20}} + R^{\frac{1}{4}} A^{\frac{7}{20}} M^{\frac{3}{10}} N^{\frac{13}{20}} + R^{\frac{1}{4}} A^{\frac{1}{2}} N^{\frac{7}{8}} \right) \\ & \ll M^\epsilon \|\alpha\| \|\nu\| \|\beta\| (AMN)^{\frac{1}{2}} R^{\frac{1}{4}} \left(1 + \frac{|\vartheta|A}{MN}\right)^{\frac{1}{4}} \left(\frac{1}{N^{\frac{1}{8}}} + \frac{R^{\frac{1}{8}} N^{\frac{1}{8}}}{M^{\frac{1}{4}}} + \frac{M^{\frac{1}{10}}}{R^{\frac{3}{20}} A^{\frac{3}{10}} N^{\frac{3}{20}}} + \frac{N^{\frac{3}{20}}}{A^{\frac{3}{20}} M^{\frac{1}{5}}} + \frac{N^{\frac{3}{8}}}{M^{\frac{1}{2}}} \right). \end{aligned}$$

4. PROOF OF THEOREM 2.3: AN IMPROVED BOUND FOR DISPERSION

Proof. Now we apply Theorem 2.1 to the work of Fouvry and Radziwiłł to prove Theorem 2.3.

Recall that

$$\Delta(\alpha, \beta, M, N, Q, a) = \sum_{\substack{q \sim Q \\ (a, q) = 1}} \left| \sum_{\substack{m \sim M \\ n \sim N \\ mn \equiv a \pmod{q}}} \alpha_m \beta_n - \frac{1}{\varphi(q)} \sum_{\substack{m \sim M \\ n \sim N \\ (mn, q) = 1}} \alpha_m \beta_n \right|.$$

Define $c_q = \pm 1$ or 0 to be such that if $(a, q) = 1$ then $c_q = \pm 1$ with the sign matching the sign of the term inside of the absolute value above, while if $(a, q) > 1$ then $c_q = 0$. In [FR], the authors bound this quantity by taking a well-chosen smooth, compactly supported function ψ with bounded derivatives and using the fact that

$$(6) \quad \sum_{m \equiv a \pmod{q}} \psi\left(\frac{m}{M}\right) = \hat{\psi}(0) \frac{M}{q} + \frac{M}{q} \sum_{0 < |h| \leq H} e\left(\frac{ah}{q}\right) \hat{\psi}\left(\frac{h}{q/M}\right) + O(M^{-1}),$$

and

$$(7) \quad \sum_{(m, q) = 1} \psi\left(\frac{m}{M}\right) = \frac{\varphi(q)}{q} \hat{\psi}(0) M + O(\tau(q)(\log 2M)^2).$$

Applying Cauchy-Schwarz, the authors bound Δ as

$$|\Delta(\alpha, \beta, M, N, Q, a)| \ll \|\alpha\| (W(Q) - 2\Re V(Q) + U(Q))^{\frac{1}{2}},$$

where

$$\begin{aligned} U(Q) &= \sum_m \psi\left(\frac{m}{M}\right) \left| \sum_{\substack{q \sim Q, n \sim N \\ (mn, q) = 1}} \frac{c_q}{\varphi(q)} \beta_n \right|^2, \\ V(Q) &= \sum_m \psi\left(\frac{m}{M}\right) \left(\sum_{\substack{q \sim Q, n \sim N \\ mn \equiv a \pmod{q}}} c_q \beta_n - \sum_{\substack{q \sim Q, n \sim N \\ (mn, q) = 1}} \frac{c_q}{\varphi(q)} \beta_n \right)^2, \\ W(Q) &= \sum_m \psi\left(\frac{m}{M}\right) \left| \sum_{\substack{q \sim Q, n \sim N \\ mn \equiv a \pmod{q}}} c_q \beta_n \right|^2. \end{aligned}$$

We use the same estimates for $U(Q)$ and $V(Q)$ as were found in the original paper.

For $W(Q)$, Fouvry and Radziwiłł apply (6) in [FR, (26) and (27)] to find that

$$W(Q) = \widetilde{W}^{MT}(Q) + \widetilde{W}^{Err1}(Q) + \widetilde{W}^{Err2}(Q) + O(MN^2Q^{-1}X^{\eta - \frac{\varepsilon}{2}} + X^{1 + \eta + \frac{\varepsilon}{2}}),$$

with each of the three terms corresponding to the three terms in (6) and where $\eta > 0$ is arbitrarily small. We will use the same estimates for $\widetilde{W}^{MT}(Q)$ and $\widetilde{W}^{Err2}(Q)$ as in the original paper and will focus our efforts on improving $\widetilde{W}^{Err1}(Q)$.

Take $D < N^{10}$ as described in the statement of Theorem 2.3. In (48) of [FR], the authors arrive at the following estimate:

$$\widetilde{W}^{Err1} \ll DC^4 L^{100} MQ^{-2} \sup_{\substack{(\alpha_1, \dots, \alpha_5) \\ (d, d_1, \delta, \delta_1, \delta_2)}} \sum_{\nu'_1 \leq 2N} \sum_{\nu_2 \leq 2N} \left| \sum_{1 \leq h \leq H} \sum_{k'_1 \leq 2Q} \sum_{k'_2 \leq 2Q} \eta_0(h) \eta_1(k'_1) \eta_2(k'_2) \right. \\ \left. \times e \left(\frac{ah\gamma \overline{dd_1}(d_1\nu'_1 - \nu_2)(\overline{\nu_2 k'_1})}{\nu'_1 k'_2} \right) \right|,$$

where the η_i are 1-bounded functions, the d 's and δ 's and γ are small parameters, and the α_i are congruence requirements for the ν 's and k 's mod γdd_1 , and

$$H = 4L^4 Q^2 M^{-1}.$$

We now translate to the notation of [BC] to the current task (the former is on the left of the arrow, the latter on the right) and apply Theorem 2.1 with

$$\vartheta \rightarrow a(d_1\nu'_1 - \nu_2), \quad a \rightarrow \underline{h}, \quad m \rightarrow \gamma dd_1 \nu_2 \underline{k'_1}, \quad n \rightarrow \underline{k_2}, \quad \nu'_1 \rightarrow R,$$

where the underlined terms are the ones that are considered to be non-fixed. This makes our translation

$$|\vartheta| \ll |a|DN + |a|N, \quad A \rightarrow H, \quad M \rightarrow \gamma dd_1 \nu_2 Q \ll NQ^{1+\epsilon}, \quad N \rightarrow Q, \quad R \rightarrow N.$$

(The authors of [FR] actually break the sums into dyadic intervals and then bound the intervals appropriately, but this step only multiplies a factor of M^ϵ to the final bound, so we will skip it.)

Then

$$\begin{aligned} W^{Err1} &\ll \mathcal{L}^{200} D^{C_5} MQ^{-2} N^2 Q H^{\frac{1}{2}} \left(1 + \frac{|a|}{MN} \right)^{\frac{1}{4}} \\ &\quad H^{\frac{1}{2}} N^{\frac{3}{4}} Q \left(\frac{1}{Q^{\frac{1}{8}}} + \frac{1}{(QN)^{\frac{1}{8}}} + \frac{1}{(NQ)^{\frac{1}{20}} H^{\frac{3}{10}}} + \frac{1}{H^{\frac{3}{20}} N^{\frac{1}{5}} Q^{\frac{1}{20}}} + \frac{1}{N^{\frac{1}{8}} Q^{\frac{1}{8}}} \right) \\ &\ll \mathcal{L}^{200} D^{C_5} MN^{\frac{11}{4}} H \left(1 + \frac{|a|}{MN} \right)^{\frac{1}{4}} \left(\frac{1}{Q^{\frac{1}{8}}} + \frac{1}{(NQ)^{\frac{1}{20}} H^{\frac{3}{10}}} + \frac{1}{H^{\frac{3}{20}} N^{\frac{1}{5}} Q^{\frac{1}{20}}} \right) \\ &\ll \mathcal{D}^{C_5} X^\epsilon \left(Q^{\frac{15}{8}} N^{\frac{11}{4}} + M^{\frac{3}{20}} Q^{\frac{27}{20}} N^{\frac{27}{10}} + M^{\frac{3}{20}} Q^{\frac{33}{20}} N^{\frac{51}{20}} \right), \end{aligned}$$

where the penultimate line eliminates the dominated $\frac{1}{(QN)^{\frac{1}{8}}}$ -terms and the last line substitutes the definition of H .

Since $N < Q$, the second term is dominated by the third one, and hence

$$(8) \quad W^{Err1} \ll \mathcal{D}^{C_5} X^\epsilon \left(Q^{\frac{15}{8}} N^{\frac{11}{4}} + M^{\frac{3}{20}} Q^{\frac{33}{20}} N^{\frac{51}{20}} \right).$$

In [FR], the authors find

$$([FR, (49)]) \quad W^{Err1} \ll \mathcal{D}^{C_5} X^\epsilon \left(Q^{\frac{15}{8}} N^{\frac{23}{8}} + Q^{\frac{33}{20}} M^{\frac{3}{10}} N^{\frac{59}{20}} \right)$$

Hence, we can replace [FR, (49)] with our new bound (8) in their Theorem 1.1 to find

$$|\Delta(\alpha, \beta, M, N, Q, a)|$$

$$\ll \|\alpha\| \left(MQ^{-1} \mathcal{E}^*(\beta, N, Q) + (\log X)^\kappa N^2 Q + (\log X)^\kappa N^2 D^{-\frac{1}{2}} M + D^C X^\epsilon \left(Q^{\frac{15}{8}} N^{\frac{11}{4}} + M^{\frac{3}{20}} Q^{\frac{33}{20}} N^{\frac{51}{20}} \right) \right)^{\frac{1}{2}}.$$

Since β is Siegel-Walfisz and α and β are k -fold divisor-bounded, we have $\mathcal{E}^*(\beta, N, Q) \ll N^2 Q \log^{-A} X$ for every $A > 0$ and thus

$$|\Delta(\alpha, \beta, M, N, Q, a)| \ll X \log^{-A} X + (\log^\eta X) N (\sqrt{MQ} + D^{-\frac{1}{2}} M) + D^{C'} X^\epsilon M^{\frac{1}{2}} Q^{\frac{15}{16}} N^{\frac{11}{8}} + Q^{\frac{33}{40}} M^{\frac{23}{40}} N^{\frac{51}{40}}.$$

□

5. PROOF OF COROLLARY 2.2: UNBALANCED CONVOLUTIONS

Beginning with the proof of (i), take

$$D = ((\log 2X)^\epsilon / 10).$$

Hence

$$X \log^{-A} X + (\log^\eta X) N (\sqrt{MQ} + D^{-\frac{1}{2}} M) \ll X \log^{-A} X.$$

By Shiu's theorem [Sh], since $|\alpha_m| \leq \tau_k(m)$,

$$||\alpha|| \ll M^{\frac{1}{2}} (\log X)^{(k^2-1)}.$$

So we require

$$(9) \quad M^{\frac{1}{2}} Q^{\frac{15}{16}} N^{\frac{11}{8}} < X^{1-\epsilon},$$

and

$$(10) \quad M^{\frac{23}{40}} Q^{\frac{33}{40}} N^{\frac{51}{40}} < X^{1-\epsilon}.$$

Recalling that $MN \leq X$, (9) will be true when

$$(11) \quad Q^{\frac{15}{16}} N^{\frac{7}{8}} < X^{\frac{1}{2}-\epsilon},$$

while (10) is true when

$$(12) \quad Q^{\frac{33}{40}} N^{\frac{28}{40}} < X^{\frac{17}{40}-\epsilon}.$$

Solving both for N gives

$$N \leq Q^{-\frac{15}{14}} X^{\frac{4}{7}-\epsilon},$$

$$N \leq Q^{-\frac{33}{28}} X^{\frac{17}{28}-\epsilon}.$$

The latter is clearly the more restrictive as long as $Q \geq \sqrt{X}$. This completes part (i) of the theorem.

For (ii), we recall a result of Fouvry [Fo2, Corollaire 1] that states that (1) holds when $1 < |a| < X/3$ and

$$Q \leq \min(\sqrt{NX}, X^{\frac{4}{7}} N^{-\frac{6}{7}}) \text{ and } N > X^\epsilon.$$

Taking $Q \leq X^{45/89-\epsilon}$, our theorem allows us to take $N \leq X^{1/89-\epsilon}$, while Fouvry's result allows $X^{1/89-\epsilon} \leq N \leq X^{7/90-\epsilon}$. Similarly, for (iii), another result of Fouvry [Fo1, Théorème 1] allows 1 to hold when $1 < |a| < \epsilon/1000$ and

$$Q \leq \min(\sqrt{NX}, X^{\frac{5}{8}} N^{-\frac{3}{4}}) \text{ and } N > X^\epsilon.$$

Our result again then gives $N \leq X^{1/89-\epsilon}$, while Fouvry's result allows $X^{1/89-\epsilon} \leq N \leq X^{101/630-\epsilon}$.

REFERENCES

- [BC] S. Bettin and V. Chandee. Trilinear forms with Kloosterman fractions. *Adv. Math.*, 328 (2018) 1234–1262.
- [DF1] W. Duke, J. Friedlander, and H. Iwaniec. Bilinear forms with Kloosterman fractions. *Invent. Math.*, 128(1):23–43, 1997.
- [Fo1] É. Fouvry, Autour du théorème de Bombieri-Vinogradov. *Acta Math.*, 152(3-4) (1984) 219–244.
- [Fo2] É. Fouvry, Théorème de Brun-Titchmarsh: application au théorème de Fermat, *Invent. Math.*, 79(2) (1985) 383–407.
- [FR] É. Fouvry and M. Radziwiłł, Level of distribution of unbalanced sequences, *Ann. Sci. Ec. Norm. Super. (4)*, 55(2) (2022), 537-568.
- [Sh] P. Shiu, A Brun-Titchmarsh theorem for multiplicative functions, *J. Reine Angew. Math.* 313 (1980), 161-170.